

Chapter 54

SD/MMC Host Controller (SDHOST)

54.1 Overview

The ESP32-P4 memory card interface controller provides a hardware interface between the Advanced Peripheral Bus (APB) and an external memory device. The memory card interface allows the ESP32-P4 to be connected to SDIO (secure digital I/O) memory cards, MMC (multimedia cards) and devices with a CE-ATA (Consumer Electronics Advanced Transport Architecture) interface. It supports two external cards (Card0 and Card1). All SD/MMC module interface signals only connect to GPIO pins via GPIO matrix.

54.2 Features

This module supports the following features:

- Two external cards
- SD memory Card specification V3.0 and V3.01, with a maximum transfer rate of DDR50
- MMC: V4.41, V4.5, and V4.51, with a maximum transfer rate of DDR50
- CE-ATA: V1.1
- 1-bit, 4-bit, and 8-bit modes

The SD/MMC controller topology is shown in Figure 54.2-1. The controller supports two peripherals, but they cannot function at the same time.

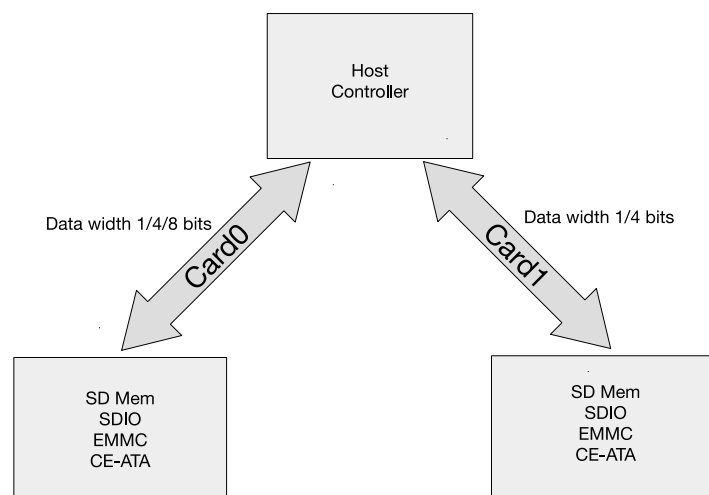


Figure 54.2-1. SD/MMC Controller Topology

54.3 SD/MMC External Interface Signals

The primary external interface signals, which enable the SD/MMC controller to communicate with an external device, are clock (sdhost_cclk_out_1.eg:card1), command (sdhost_ccmd_out_1), and data signals (sdhost_cdata_in_1[7:0]/sdhost_cdata_out_1[7:0]). Additional signals include the card interrupt, card detect, and write protect signals. The direction of each signal is shown in Figure 54.3-1. The direction and description of each pin are listed in Table 54.3-1.

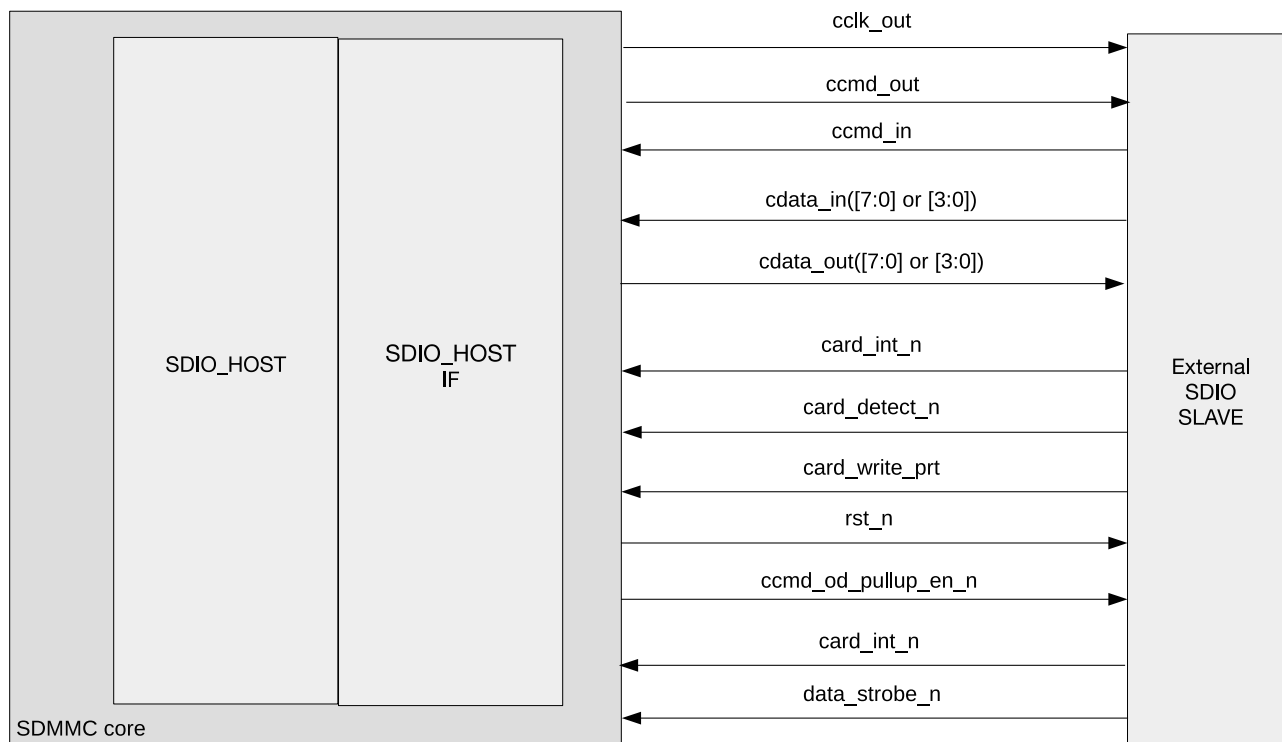


Figure 54.3-1. SD/MMC Controller External Interface Signals

Table 54.3-1. SD/MMC Signal Description

Signal	Direction	Description
sdhost_cclk_out	Output	Clock signals for slave device
sdhost_ccmd	Duplex	Duplex command/response lines
sdhost_cdata	Duplex	Duplex data read/write lines
sdhost_card_detect_n	Input	Card detection input line
sdhost_card_write_prt	Input	Card write protection status input

54.4 Functional Description

54.4.1 SD/MMC Host Controller Architecture

The SD/MMC host controller consists of two main functional blocks, as shown in Figure 54.4-1:

- Bus Interface Unit (BIU): Provides the APB interface for registers, access to RAM data, and DMA data read and write operations.

- Card Interface Unit (CIU): Handles external memory card interface protocols and provides clock control.

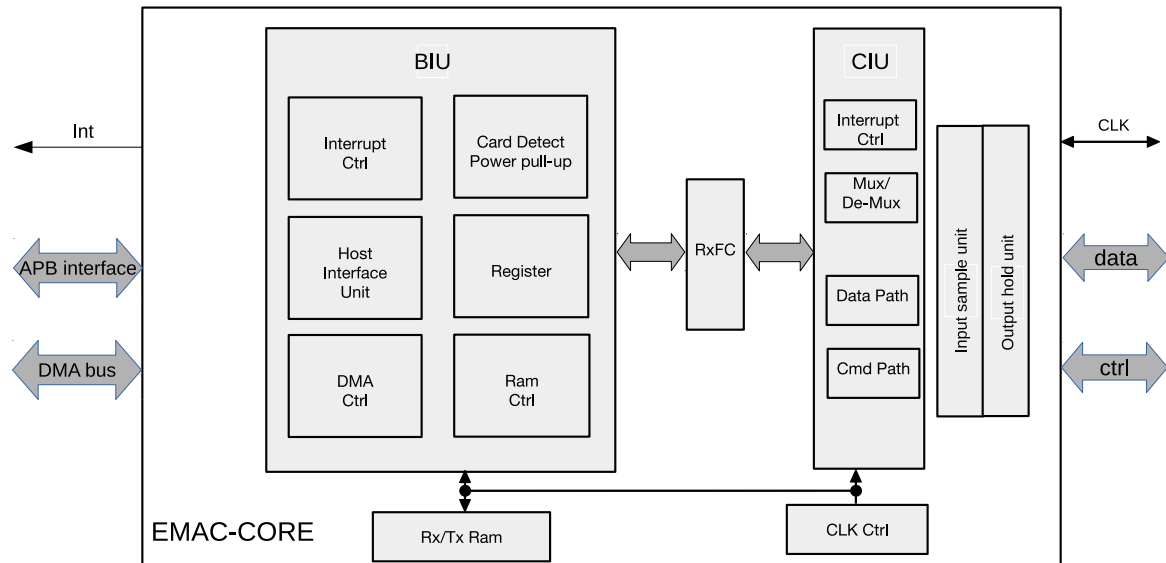


Figure 54.4-1. SDIO Host Block Diagram

54.4.1.1 Bus Interface Unit (BIU)

The BIU provides access to registers and RAM data through the Host Interface Unit (HIU). Additionally, it provides a method to access to memory data through a DMA interface. Figure 54.4-1 illustrates the internal components of the BIU.

The BIU provides the following functions:

- Host interface
- DMA interface
- Interrupt control
- Register access
- FIFO access
- Power-up/pull-up control and card detection

54.4.1.2 Card Interface Unit (CIU)

The CIU module implements the dedicated card slot specifications. Within the CIU, the command path control unit and data path control unit are used to interface with the command and data ports, respectively, of the SD/MMC/CE-ATA cards. The CIU also provides clock control. Figure 54.4-1 illustrates the internal structure of the CIU, which consists of the following primary functional blocks:

- Command path
- Data path
- SDIO interrupt control
- Clock control
- MUX (multiplexer) and De-MUX (de-multiplexer) unit

54.4.2 Command Path

The command path performs the following functions:

- Configures clock parameters
- Configures card command parameters
- Sends commands to card bus (sdhost_ccmd_out line)
- Receives responses from card bus (sdhost_ccmd_in line)
- Sends responses to BIU
- Drives the P-bit on the command line

The command path state machine is shown in Figure 54.4-2.

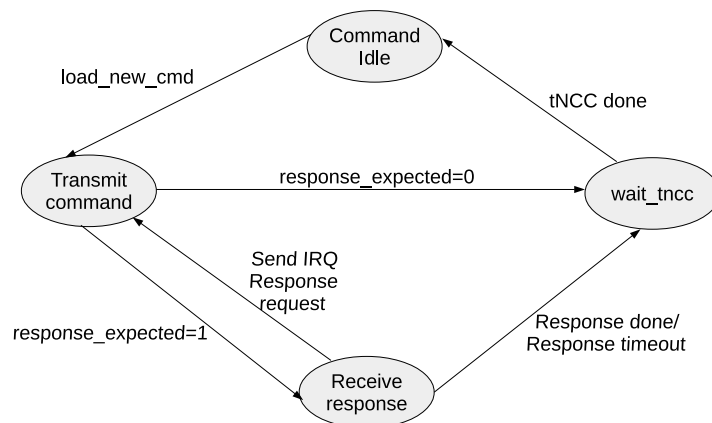


Figure 54.4-2. Command Path State Machine

54.4.3 Data Path

The data path block pops RAM data and transmits them on sdhost_cdata_out during a write-data transfer, or it receives data on sdhost_cdata_in and pushes them into RAM during a read-data transfer. The data path loads new data parameters, i.e., expected data, read/write data transfer, stream/block transfer, block size, byte count, card type, timeout registers, etc., whenever a data transfer command is not in progress.

If the [SDHOST_DATA_EXPECTED](#) bit is set in the [SDHOST_CMD_REG](#) register, the new command is a data transfer command and the data path starts one of the following operations:

- Transmitting data if the [SDHOST_READ_WRITE](#) bit is 1
- Receiving data if the [SDHOST_READ_WRITE](#) bit is 0

54.4.3.1 Data Transmit

The module starts data transmission two clock cycles after a response for the data write command is received. This occurs even if the command path detects a response error or a cyclic redundancy check (CRC) error in a response. If no response is received from the card until the response timeout, no data will be transmitted. Depending on the value of the [SDHOST_TRANSFER_MODE](#) bit in the [SDHOST_CMD_REG](#) register, the data transmit state machine adds data to the card's data bus in a stream or in block(s). The data transmit state machine is shown in Figure 54.4-3.

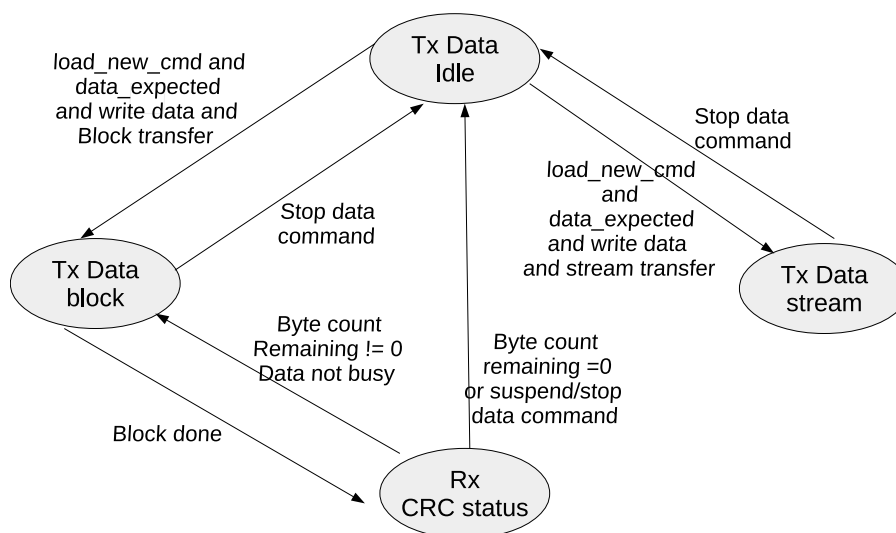


Figure 54.4-3. Data Transmit State Machine

54.4.3.2 Data Receive

The module receives data two clock cycles after the end bit of a data read command, even if the command path detects a response error or a CRC error. If no response is received from the card and a response timeout occurs, the BIU does not receive a signal about the completion of the data transfer. If the command sent by the CIU is an illegal operation for the card, it would prevent the card from starting a read-data transfer, and the BIU will not receive a signal about the completion of the data transfer.

If no data is received by the data timeout, the data path signals a data timeout to the BIU, which marks an end to the data transfer. Based on the value of the `SDHOST_TRANSFER_MODE` bit in the `SDHOST_CMD_REG` register, the data receive state machine gets data from the card's data bus in a stream or block(s). The data receive state machine is shown in Figure 54.4-4.

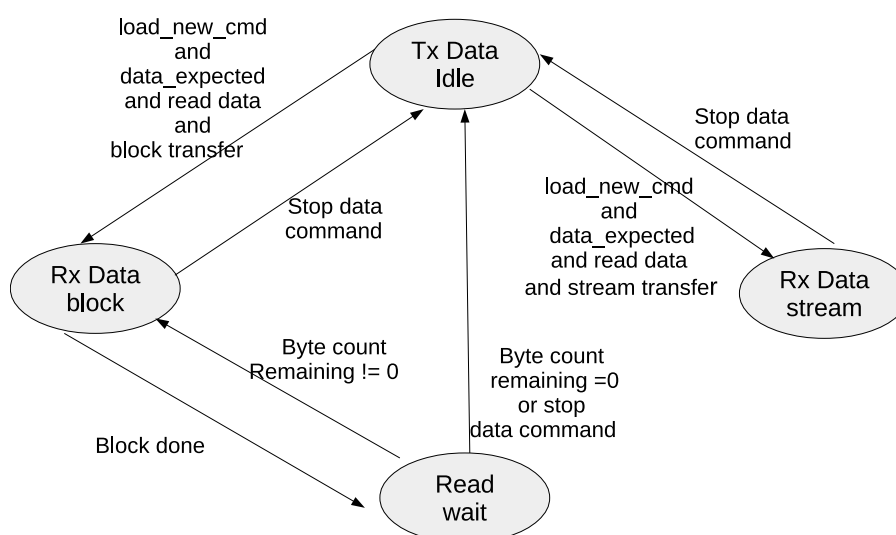


Figure 54.4-4. Data Receive State Machine

54.5 Software Restrictions for CIU Operations

- Only one card at a time can be selected to execute a command or data transfer. For example, when data are being transferred to or from a card, a new command must not be issued to another card. A new command, however, can be issued to the same card, allowing it to read the device status or stop the transfer.
- Only one data transfer command can be issued at a time.
- During an open-ended card write operation, if the card clock is stopped because RAM is empty, the software must first fill RAM with data, and start the card clock. Only then can it issue a stop/abort command to the card.
- During an SDIO/Combo card transfer, if the card function is suspended and the software wants to resume the suspended transfer, it must first reset and then release RAM by configuring the [SDHOST_FIFO_RESET](#) bit, and then issue the resume command as if it were a new data transfer command.
- When issuing card reset commands (CMD0, CMD15 or CMD52_reset) while a card data transfer is in progress, the software must set the [SDHOST_STOP_ABORT_CMD](#) bit in the [SDHOST_CMD_REG](#) register, so that the CIU can stop the data transfer after issuing the card reset command.
- When the controlling bit for end bit error is set in the [SDHOST_RINTSTS_REG](#) register, the CIU cannot control SDIO interrupts. In such a case, the software must ignore SDIO interrupts and issue a stop/abort command to the card, so that the card stops transmitting data.
- If the card clock is stopped because RAM is full during a card read, the software must read at least two RAM addresses to restart the card clock.
- Only one CE-ATA device at a time can be selected for a command or data transfer. For example, when data are transferred from a CE-ATA device, a new command must not be sent to another CE-ATA device.
- If the CE-ATA device interrupts are enabled (nIEN=0), a new SDHOST_RW_BLK command must not be sent to the same device if the execution of a SDHOST_RW_BLK command is already in progress. Only the Command Completion Signal Disable (CCSD) command can be sent while waiting for the Command Completion Signal (CCS).
- If, however, the CE-ATA device interrupts are disabled (nIEN=1), a new command can be issued to the same device, allowing it to read status information.
- Open-ended transfers are not supported for CE-ATA devices.
- The sdhost_send_auto_stop signal is not supported (software must not set the [SDHOST_SEND_AUTO_STOP](#) bit) for CE-ATA transfers.

After configuring the command start bit to 1, the values of the following registers cannot be changed before a command has been issued:

- CMD - command
- CMDARG - command argument
- BYTCNT - byte count
- BLKSIZ - block size
- CLKDIV - clock divider

- CKLENA - clock enable
- CLKSRC - clock source
- TMOUT - timeout
- CTYPE - card type

54.6 RAM for Receiving and Transmitting Data

The submodule RAM is a buffer area for transmitting and receiving data. It can be divided into two units: one is for transmitting data, and the other is for receiving data. The process of transmitting and receiving data can be achieved by the CPU and DMA for reading and writing. The latter method is described in detail in Section [54.8](#).

54.6.1 TX RAM Module

There are two ways to enable a write operation: CPU and DMA write.

For CPU write operations to TX RAM, write the data directly into the [SDHOST_BUFFIFO_REG](#) register via the APB interface. The corresponding FIFO width is 32 bits and the depth is 512.

For DMA write operations to TX RAM, set up DMA linked list descriptors as described in Section [54.8](#).

54.6.2 RX RAM Module

There are two ways to enable a read operation: CPU and DMA read.

For CPU read operations from RX RAM, read the data directly from the [SDHOST_BUFFIFO_REG](#) register via the APB interface. The corresponding FIFO width is 32 bits and the depth is 512.

For DMA read operations from RX RAM, set up DMA linked list descriptors as described in Section [54.8](#).

54.7 DMA Linked List

Each unit in the linked list structure consists of two parts: a descriptor and a data buffer. In other words, each descriptor points to a unique data buffer and to the next descriptor in the linked list. Figure [54.7-1](#) shows the linked list.

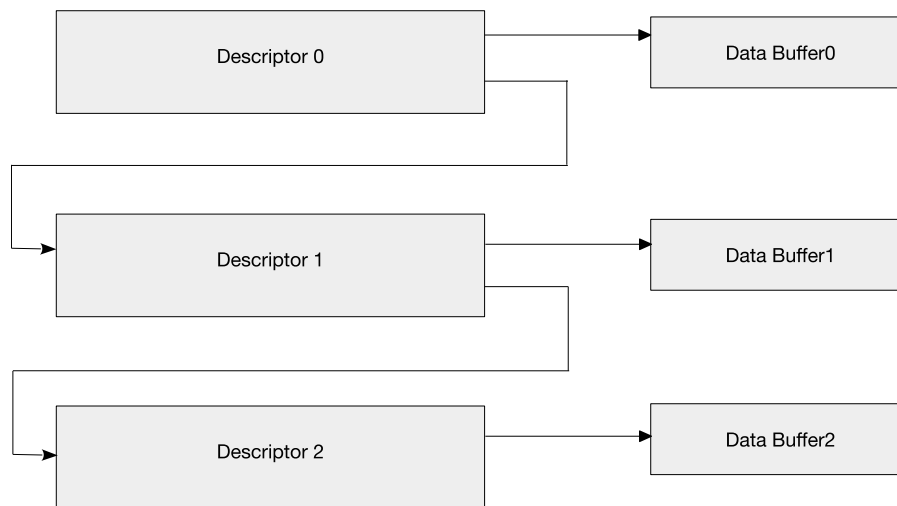


Figure 54.7-1. Linked List Structure

54.8 DMA Descriptor Format

Each descriptor consists of four words as shown in 54.8-1. Table 54.8-1, Table 54.8-2, Table 54.8-3, Table 54.8-4 provide the descriptions of these four words.

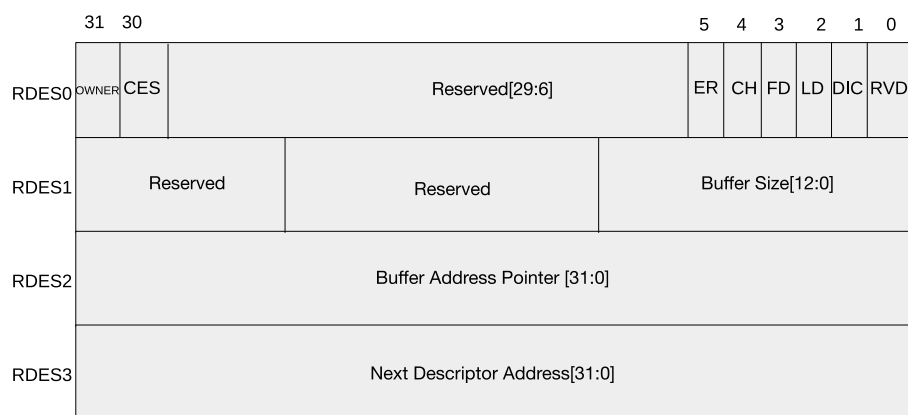


Figure 54.8-1. Descriptor Format

The DESO field contains control and status information.

Table 54.8-1. DESO Descriptor Field

Bits	Name	Description
31	OWNER	When set to 1, this bit indicates that the descriptor is owned by the DMA Controller. When reset to 0, it indicates that the descriptor is owned by the host. The DMA clears this bit when it completes the data transfer.

Bits	Name	Description
30	CES (Card Error Summary)	This bit indicates the status of the card's read/write operations. This bit is the logical OR of the following bits in the SDHOST_RINTSTS_REG register. • EBE: End Bit Error • RTO: Response Timeout • RCRC: Response CRC • SBE: Start Bit Error • DRTO: Data Read Timeout • DCRC: Data CRC for Receive • RE: Response Error
29:6	Reserved	Reserved
5	ER (End of Ring)	When set to 1, this bit indicates that the linked list has reached its final descriptor. The DMA Controller then returns to the base address of the linked list, creating a linked list ring.
4	CH (Second Address Chained)	When set to 1, this bit indicates that the second address in the descriptor is the next descriptor address. When this bit is set to 1, BS2 (DES1[25:13]) must be all zeros.
3	FD (First Descriptor)	When set to 1, this bit indicates that this descriptor contains the first buffer of the data. If the size of the first buffer is 0, the next descriptor contains the beginning of the data.
2	LD (Last Descriptor)	This bit is associated with the last data block of a DMA transfer. When set to 1, the bit indicates that the buffers pointed by this descriptor are the last buffers of the data. After this descriptor is completed, the remaining byte count is 0. In other words, after the descriptor with the LD bit set is completed, the remaining byte count should be 0.
1	DIC (Disable Interrupt on Completion)	When set to 1, this bit prevents the setting of the TI/RI bit of the DMA Status Register (SDHOST_IDSTS_REG) for the data that ends in the buffer pointed by this descriptor.
0	Reserved	Reserved

The DES1 field contains the buffer size.

Table 54.8-2. DES1 Descriptor Field

Bits	Name	Description
31:26	Reserved	Reserved
25:13	Reserved	Reserved

Bits	Name	Description
12:0	BS (Buffer Size)	These bits indicate the size of the data buffer byte size, which must be a multiple of four. When the buffer size is not a multiple of four, the resulting behavior is undefined. This field must not be zero.

The DES2 field contains the address pointer to the data buffer.

Table 54.8-3. DES2 Descriptor Field

Bits	Name	Description
31:0	Buffer Address Pointer	These bits indicate the address of the data buffer. The buffer address must be word-aligned.

The DES3 field contains the address pointer to the next descriptor if the present descriptor is not the last one in a linked list structure.

Table 54.8-4. DES3 Descriptor Field

Bits	Name	Description
31:0	Next Descriptor Address	If CH (DESO[4]) is set to 1, these bits contains the pointer to the next descriptor. If this is not the last descriptor in a linked list structure, bits 1 and 0 must be zero, i.e., DES3[1:0] = 0.

54.9 Programming Procedures

54.9.1 Initializing Registers

To initialize registers, perform the following steps:

1. Write to the control register [SDHOST_CTRL_REG](#), the raw interrupt register [SDHOST_RINTSTS_REG](#), and SDIO interrupt mask register [SDHOST_INTMASK_REG](#) to clear pending interrupts and configure controlling and interrupt parameters.
2. Write clock divider configuration register [SDHOST_CLKDIV_REG](#), clock source selection register [SDHOST_CLKSRC_REG](#), and clock enable register [SDHOST_CLKENA_REG](#) to configure the card clock.
3. Write other configuration registers based on card parameters. For example, configure
 - card bus width ([SDHOST_CTYPE_REG](#))
 - user ID ([SDHOST_USRID_REG](#))
 - UHS mode voltage and DDR ([SDHOST_UHS_REG](#))
 - EMMC mode start bit ([SDHOST_EMMCDDR_REG](#))
 - SDIO mode ([SDHOST_CLK_EDGE_SEL_REG](#))
 - timeout value ([SDHOST_TMOUT_REG](#))

- debounce filter ([SDHOST_DEBNCE_REG](#))
- internal phase shift ([SDHOST_ENSHIFT_REG](#))

You can reset the card with [SDHOST_RST_N_REG](#).

54.9.2 Sending Commands

To send commands, perform the following steps:

1. Write the command argument register [SDHOST_CMDARG_REG](#) with the appropriate command argument parameter.
2. For data transfer, write the data size register [SDHOST_BYTCNT_REG](#), the block size register [SDHOST_BLKSIZE_REG](#) register, and the card threshold control register [SDHOST_CARDTHRCTL_REG](#) to configure data transfer parameters.
3. Data transfer can be achieved by DMA or FIFO, and DMA is preferable.
 - To transmit and receive data via DMA, configure the DMA Controller according to Section [54.9.3](#), Section [54.9.4](#), and Section [54.9.5](#).
 - To transmit and receive data via FIFO, configure the FIFO threshold via FIFO configuration register [SDHOST_FIFOTH_REG](#). For writing to a card, software should write data to the FIFO via [SDHOST_BUFFIFO_REG](#) continuously during command execution.
4. Configure and start the command via the [SDHOST_CMD_REG](#).
5. Wait until the [SDHOST_START_CMD](#) bit in the [SDHOST_CMD_REG](#) register is cleared by hardware upon command reception. When hardware cannot receive the command, it will set the 12th bit HLE in the [SDHOST_RINTSTS_REG](#) register to 1.
6. Read the [SDHOST_RINTSTS_REG](#) register or enable interrupts to check command execution status. If needed, read the [SDHOST_RESP \$n\$ _REG](#) (n ranges from 0 ~ 3) register to get command response from the card.
7. If data is read from the card via FIFO, software can get the data returned by the card via reading the [SDHOST_BUFFIFO_REG](#) register.

Please refer to the card specification for the command format, and configure register field values depending on the command format.

54.9.3 Initializing DMA

To initialize the DMA Controller, perform the following steps:

1. Write to the DMA bus mode register ([SDHOST_BMOD_REG](#)) to configure the host bus's access parameters.
2. Write to the DMA interrupt enable register ([SDHOST_IDINTEN_REG](#)) to mask unnecessary interrupt causes.
3. Create either a transmit or receive linked list. Then write the starting address of the first descriptor in the linked list to the DMA Controller's linked list base address register ([SDHOST_DBADDR_REG](#)).
4. The DMA Controller engine proceeds to get descriptors from the linked list.

54.9.4 Initializing DMA Transmission

To initialize DMA transmission, perform the following steps:

1. The host sets up the descriptor fields (DESO-DES3) for transmission, and sets the OWNER bit (DESO[31]) to 1. The host also prepares the data buffer.
2. The host writes the write data command to the command register [SDHOST_CMD_REG](#).
3. The host sets the required transmit threshold via the [SDHOST_TX_WMARK](#) field in the [SDHOST_FIFOTH_REG](#) register.
4. The DMA Controller engine fetches the descriptor and checks the OWNER bit. If the OWNER bit is not set, it means that the host owns the descriptor. In this case, the DMA Controller enters a suspend state and asserts the Descriptor Unable interrupt via the [SDHOST_IDSTS_REG](#) register. The host then needs to release the DMA Controller by writing any value to [SDHOST_PLDMND_REG](#).
5. The DMA Controller waits for the Command Done (CMDD) bit in the [SDHOST_RINTSTS_REG](#) register to be set to 1 with no errors from BIU, which indicates that a transfer has completed.
6. The DMA Controller engine waits for a DMA interface request from BIU. This request is generated based on the configured transmit threshold value. For the last byte of data that cannot be accessed using a burst, a single transfer is performed on the AHB bus.
7. The DMA Controller fetches the transmit data from the data buffer in the host memory and transfers them to RAM in preparation for transmission to the card.
8. When data span across multiple descriptors, the DMA Controller fetches the next descriptor and continues with its operation with the next descriptor. The Last Descriptor bit indicates whether the data span multiple descriptors or not.
9. When data transmission is complete, the status information is updated in the [SDHOST_IDSTS_REG](#) register by setting the [SDHOST_IDSTS_TI](#) bit to 1 if it has already been enabled. Also, the OWNER bit is cleared by the DMA Controller by updating the DESO field.

54.9.5 Initializing DMA Reception

To initialize DMA reception, perform the following steps:

1. The host sets up the descriptor fields (DESO-DES3) for reception, and sets the OWNER bit (DESO[31]) to 1.
2. The host writes the read data command in the [SDHOST_CMD_REG](#) register in BIU.
3. The host sets the required receive threshold level via the [SDHOST_RX_WMARK](#) field in the [SDHOST_FIFOTH_REG](#) register.
4. The DMA Controller engine fetches the descriptor and checks the OWNER bit. If the OWNER bit is not set, it means that the host owns the descriptor. In this case, the DMA enters suspend state and asserts the Descriptor Unable interrupt via the [SDHOST_IDSTS_REG](#) register. The host then needs to release the DMA Controller by writing any value to [SDHOST_PLDMND_REG](#).
5. The DMA Controller waits for the Command Done (CMDD) bit in the [SDHOST_RINTSTS_REG](#) register to be set to 1 with no errors from BIU, which indicates that a reception has been completed.

6. The DMA Controller engine waits for a DMA interface request from BIU. This request is generated based on the configured receive threshold value. For the last byte of data that cannot be accessed using a burst, a single transfer is performed on the AHB bus.
7. The DMA Controller fetches the data from RAM and transfers them to the host memory.
8. When data span across multiple descriptors, the DMA Controller fetches the next descriptor and continues with its operation with the next descriptor. The Last Descriptor bit indicates whether the data span multiple descriptors or not.
9. When data reception is complete, the status information is updated in the [SDHOST_IDSTS_REG](#) register by setting the [SDHOST_IDSTS_RI](#) bit to 1, if it has already been enabled. Also, the OWNER bit is cleared by the DMA Controller by updating the DESO field.

54.10 Clock Phase Selection

The phase of the SD/MMC Host Controller clock source is configurable, so as to adjust the setup time sequence for the input or output data signals.

The clock source of the SD/MMC Host Controller can be a high-performance clock at a high frequency or a low-power clock at a low frequency. When the clock source is a high-performance clock, the internal signal clock phase, output signal driving clock phase, and input signal sampling clock phase can be configured via the [SDHOST_DLL_CLK_CONF_REG](#), in the unit of 1/64 clock source period. When the clock source is a low-power clock, the internal signal clock phase, output signal driving clock phase, and input signal sampling clock phase can be configured via the system clock register. The phase options are 0 degrees, 90 degrees, 180 degrees, and 270 degrees, or in other words the unit is 1/4 clock source period.

For details, please see Chapter [10 Reset and Clock](#).

54.11 Interrupt

The SD/MMC Host Controller can generate interrupt signal SDIO_HOST_INTR triggered by the following interrupt sources, and send the interrupt signals to the [Interrupt Matrix](#).

- SDIO_INT: Interrupts from SDIO cards
- EBE: Triggered when an error in end bit occurs, or no data CRC received
- ACD: Triggered when an automatically sent command has been executed
- SBE/BCI: Triggered when an error in the start bit occurs
- HLE: Triggered when a write error occurs during hardware-lock period
- FRUN: Triggered when the FIFO is empty or full
- HTO: Triggered when the host does not fill data before timeout period
- DRTO: Triggered when a data read operation times out
- RTO: Triggered when card response times out
- DCRC: Triggered when data CRC error occurs
- RCRC: Triggered when a card response CRC error occurs

- RXDR: Triggered during read operations from card when FIFO level is greater than Receive-Threshold level
- TXDR: Triggered during write operations to card when FIFO level reaches less than or equal to Transmit-Threshold level.
- DTO: Triggered when data transfer completes
- CMDD: Triggered when a command has been executed
- RE: Triggered when response error occurs
- CD: Triggered when a card is detected
- IDSTS_RI: Triggered when data pointed by a DMA descriptor list have been received
- IDSTS_TI: Triggered when data pointed by a DMA descriptor list have been transmitted
- IDSTS_FBE: Triggered when a fatal bus error occurs and the host is stopped during data transmission or reception. When triggered, the DMA disables all bus access.
- IDSTS_DU: Triggered when a DMA descriptor list is not available because the OWNER bit is 0 (DESO [31] = 0)

The [SDHOST_RINTSTS_REG](#) and [SDHOST_IDSTS_REG](#) register contains all the bits that might cause an interrupt. The [SDHOST_INTMASK_REG](#) and [SDHOST_IDINTEN_REG](#) register contains an enable bit for each of the events that can cause an interrupt.

There are three interrupt summary bits in the [SDHOST_IDSTS_REG](#) register: the card error interrupt summary bit (bit 5 [SDHOST_IDSTS_CES](#)), the normal interrupt summary bit (bit 8 [SDHOST_IDSTS_NIS](#)), and the abnormal interrupt summary bit (bit 9 [SDHOST_IDSTS_AIS](#)). Interrupts are cleared by writing 1 to the corresponding bit. When all the enabled interrupts within a group are cleared, the corresponding summary bit is also cleared.

Interrupts are not queued. If another interrupt event occurs before the driver has responded to the previous interrupt, no additional interrupts are generated. For example, the [SDHOST_IDSTS_RI](#) indicates that one or more data were transferred to the host buffer.

An interrupt is generated only once for simultaneous multiple events. The driver must scan the [SDHOST_RINTSTS_REG](#) and [SDHOST_IDSTS_REG](#) register for the interrupt cause.

Note:

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

54.12 Register Summary

The addresses in this section are relative to the SD/MMC Host Controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

Name	Description	Address	Access
Configuration registers			
SDHOST_CTRL_REG	Control register	0x0000	R/W
SDHOST_CLKDIV_REG	Clock divider configuration	0x0008	R/W
SDHOST_CLKSRC_REG	Clock source selection	0x000C	R/W
SDHOST_CLKENA_REG	Clock enable register	0x0010	R/W
SDHOST_TMOUT_REG	Data and response timeout configuration	0x0014	R/W
SDHOST_CTYPE_REG	Card bus width configuration	0x0018	R/W
SDHOST_BLKSIZE_REG	Card data block size configuration	0x001C	R/W
SDHOST_BYTCNT_REG	Data transfer length configuration	0x0020	R/W
SDHOST_CMDARG_REG	Command argument data	0x0028	R/W
SDHOST_CMD_REG	Command and boot configuration	0x002C	R/W
SDHOST_FIFOTH_REG	FIFO configuration	0x004C	R/W
SDHOST_DEBNCE_REG	Debounce filter time configuration	0x0064	R/W
SDHOST_USRID_REG	User ID (scratchpad)	0x0068	R/W
SDHOST_VERID_REG	Version ID (scratchpad) register	0x006C	RO
SDHOST_HCON_REG	Hardware feature register	0x0070	RO
SDHOST_UHS_REG	UHS-1	0x0074	R/W
SDHOST_RST_N_REG	Card reset	0x0078	R/W
SDHOST_BMOD_REG	Burst mode transfer configuration	0x0080	R/W
SDHOST_PLDMND_REG	Poll demand configuration	0x0084	WO
SDHOST_DBADDR_REG	Linked-list base address	0x0088	R/W
SDHOST_CARDTHRCTL_REG	Card Threshold Control	0x0100	R/W
SDHOST_EMMCDDR_REG	eMMC DDR register	0x010C	R/W
SDHOST_ENSHIFT_REG	Enable Phase Shift	0x0110	R/W
SDHOST_CLK_EDGE_SEL_REG	SDIO control register	0x0800	R/W
SDHOST_DLL_CLK_CONF_REG	SDIO DLL clock control	0x0808	R/W
Interrupt registers			
SDHOST_INTMASK_REG	SDIO interrupt mask register	0x0024	R/W
SDHOST_MINTSTS_REG	Masked interrupt status register	0x0040	RO
SDHOST_RINTSTS_REG	Raw interrupt register	0x0044	R/W1C
SDHOST_IDSTS_REG	DMA interrupt register	0x008C	R/W1C
SDHOST_IDINTEN_REG	DMA interrupt enable register	0x0090	R/W
Status registers			
SDHOST_RESP0_REG	Response data	0x0030	RO
SDHOST_RESP1_REG	Long response data	0x0034	RO
SDHOST_RESP2_REG	Long response data	0x0038	RO
SDHOST_RESP3_REG	Long response data	0x003C	RO

Name	Description	Address	Access
SDHOST_STATUS_REG	SD/MMC status	0x0048	RO
SDHOST_CDETECT_REG	Card detect	0x0050	RO
SDHOST_WRTPRT_REG	Card write protection (WP) status	0x0054	RO
SDHOST_TCBCNT_REG	Transferred byte count	0x005C	RO
SDHOST_TBBCNT_REG	Transferred byte count	0x0060	RO
SDHOST_DSCADDR_REG	Host descriptor address pointer	0x0094	RO
SDHOST_BUFADDR_REG	Host buffer address pointer	0x0098	RO
Transmit and receive data registers			
SDHOST_BUFFIFO_REG	Software write and read transmit data by FIFO	0x0200	R/W

54.13 Registers

The addresses in this section are relative to the SD/MMC Host Controller base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

Register 54.1. SDHOST_CTRL_REG (0x0000)

(reserved)						SDHOST_USE_INTERNAL_DMAC										(reserved)						SDHOST_CEATA_DEVICE_INTERRUPT_STATUS SDHOST_SEND_AUTO_STOP_CCSD SDHOST_SEND_ABORT_READ_DATA SDHOST_SEND_IRQ_RESPONSE (reserved) (reserved) SDHOST_DMA_RESET SDHOST_FIFO_RESET SDHOST_CONTROLLER_RESET									
31						26	25	24								12	11	10	9	8	7	6	5	4	3	2	1	0	Reset		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

SDHOST_CONTROLLER_RESET Configures whether to reset the controller. This bit is auto-cleared after controller reset.

0: No effect

1: Reset

(R/W)

SDHOST_FIFO_RESET Configures whether to reset FIFO. This bit is auto-cleared after FIFO reset.

0: No effect

1: Reset

(R/W)

SDHOST_DMA_RESET Configures whether to reset DMA interface. This bit is auto-cleared after DMA reset.

0: No effect

1: Reset

(R/W)

SDHOST_INT_ENABLE Write 1 to enable global interrupt port. (R/W)

SDHOST_READ_WAIT Configures whether to send read-wait to SDIO cards.

0: Clear read wait

1: Assert read wait

(R/W)

Continued on the next page...

Register 54.1. CTRL_REG (0x0000)

Continued from the previous page...

SDHOST_SEND_IRQ_RESPONSE Configures whether to send auto interrupt request (IRQ) response.

This bit automatically clears once a response is sent.

To wait for MMC card interrupts, the host issues a CMD40 command and waits for an interrupt response from MMC card(s). In the meantime, if the host wants SD/MMC to exit waiting for the interrupt state, it can set this bit, at which time SD/MMC command state-machine sends a CMD40 response on the bus and returns to the idle state.

0: No effect

1: Send auto IRQ response

(R/W)

SDHOST_ABORT_READ_DATA Configures whether to abort read data. This bit is automatically cleared once the data state machine is reset to idle.

After a suspend-command is issued during a read-operation, software polls the card to find when the suspend-event occurred. Once the suspend-event has occurred, software sets the bit which will reset the data state machine that is waiting for the next block of data.

0: No effect

1: Abort read data

(R/W)

SDHOST_SEND_CCSD Configures whether to send Command Completion Signal Disable (CCSD) to CE-ATA device. Once the CCSD pattern is sent to the device, SD/MMC automatically clears the SDHOST_SEND_CCSD bit. It also sets the Command Done (CD) bit in the SDHOST_RINTSTS_REG register.

Software sets this bit only if the current command is expecting CCS (that is, RW_BLK), and if interrupts are enabled for the CE-ATA device.

0: No effect

1: Send CCSD to CE-ATA device

(R/W)

SDHOST_SEND_AUTO_STOP_CCSD Configures whether to send an internally-generated STOP command (CMD12) to the CE-ATA device. After sending the CCSD, SD/MMC automatically clears the SDHOST_SEND_AUTO_STOP_CCSD bit. After sending this internally-generated STOP command, the Auto Command Done (ACD) bit in SDHOST_RINTSTS_REG is set.

Always Set SDHOST_SEND_AUTO_STOP_CCSD and SDHOST_SEND_CCSD bits together, SDHOST_SEND_AUTO_STOP_CCSD should not be set independently of SDHOST_SEND_CCSD.

0: No effect

1: Send internally-generated STOP command (CMD12) to the CE-ATA device

(R/W)

Continued on the next page...

Register 54.1. CTRL_REG (0x0000)

Continued from the previous page...

SDHOST_CEATA_DEVICE_INTERRUPT_STATUS Write 1 to enable interrupts in CE-ATA device (nIEN = 0 in ATE control register).

Software should appropriately write to this bit after the power-on reset or any other reset to the CE-ATA device. After reset, the CE-ATA device's interrupt is usually disabled (nIEN = 1). If the host enables the CE-ATA device's interrupt, then software should set this bit. (R/W)

SDHOST_USE_INTERNAL_DMAC Configures whether to use DMA for data transfer.

0: Not use

1: Use

(R/W)

Register 54.2. SDHOST_CLKDIV_REG (0x0008)

<i>SDHOST_CLK_DIVIDER3</i>								<i>SDHOST_CLK_DIVIDER2</i>								<i>SDHOST_CLK_DIVIDER1</i>								<i>SDHOST_CLK_DIVIDER0</i>											
31								24	23								16	15								8	7								0
0x0								0x0								0x0								0x0								Reset			

SDHOST_CLK_DIVIDER0 Configures clock divider0 value. Clock divisor is 2^n , where $n = 0$ bypasses the divider (divisor of 1). For example, a value of 1 means divided by $2^1 = 2$, a value of 0xFF means divided by $2^{255} = 510$, and so on. (R/W)

SDHOST_CLK_DIVIDER1 Configures clock divider1 value. Clock divisor is 2^n , where $n = 0$ bypasses the divider (divisor of 1). For example, a value of 1 means divided by $2^1 = 2$, a value of 0xFF means divided by $2^{255} = 510$, and so on. (R/W)

SDHOST_CLK_DIVIDER2 Configures clock divider2 value. Clock divisor is 2^n , where $n = 0$ bypasses the divider (divisor of 1). For example, a value of 1 means divided by $2^1 = 2$, a value of 0xFF means divided by $2^{255} = 510$, and so on. (R/W)

SDHOST_CLK_DIVIDER3 Configures clock divider3 value. Clock divisor is 2^n , where $n = 0$ bypasses the divider (divisor of 1). For example, a value of 1 means divided by $2^1 = 2$, a value of 0xFF means divided by $2^{255} = 510$, and so on. (R/W)

Register 54.3. SDHOST_CLKSRC_REG (0x000C)

Diagram of the SDHOST_CLKSRC_REG register. The register is 32 bits wide. Bits 31 down to 4 are labeled "(reserved)". Bits 3 down to 0 are labeled "0x0". The register is labeled "SDHOST_CLKSRC_REG" and "Reset".

SDHOST_CLKSRC_REG Configures clock divider source for two SD cards. Each card has two bits assigned to it. For example, bit[1:0] are assigned to card 0, and bit[3:2] are assigned to card 1. Card 1/0 maps and internally routes clock divider[0:3] outputs to cclk_out[1:0] pins, depending on bit value. For every card's clock divider source:

0x0: Clock divider 0

0x1: Clock divider 1

0x2: Clock divider 2

0x3: Clock divider 3

(R/W)

Register 54.4. SDHOST_CLKENA_REG (0x0010)

[illegible]

SDHOST_CCLK_ENABLE Configures whether to enable two SD card clocks or one MMC card clock.

One bit per card. For every card's bit:

0: Clock disabled

1: Clock enabled

(R/W)

SDHOST_LP_ENABLE Configures whether to disable the clock when the card is in an IDLE state.

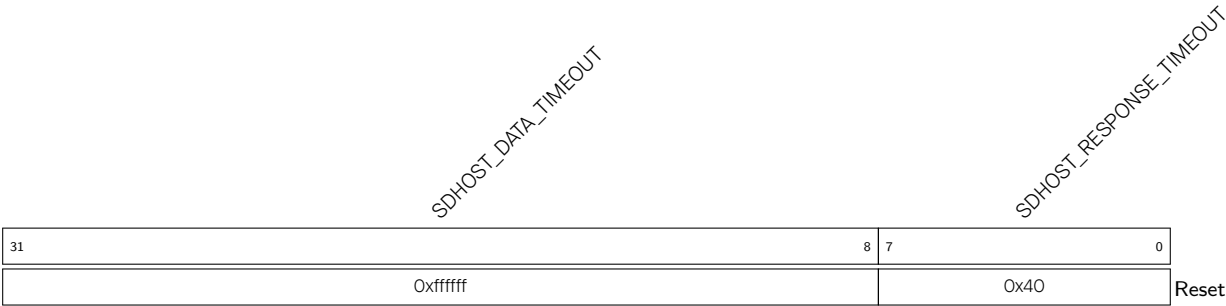
One bit per card. For every card's bit:

0: Clock disabled

1: Clock enabled

(R/W)

Register 54.5. SDHOST_TMOUT_REG (0x0014)

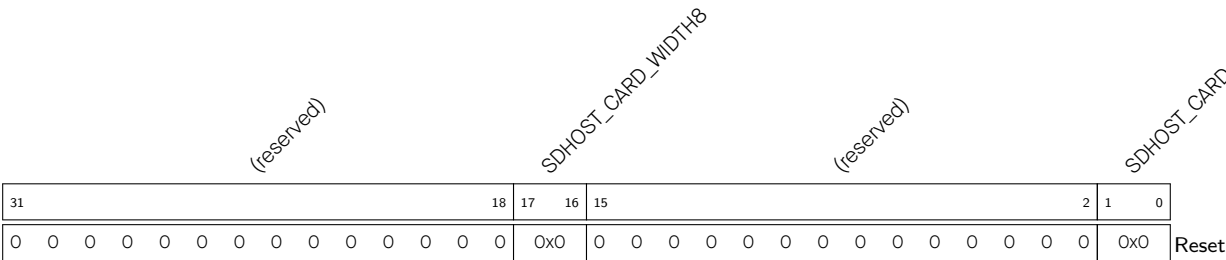


SDHOST_RESPONSE_TIMEOUT Configures response timeout value. Measurement unit: Card output clock period. (R/W)

SDHOST_DATA_TIMEOUT Configures card data read timeout value. This value is also used for data starvation by host timeout. The timeout counter is started only after the card clock is stopped. Measurement unit: Card output clock period.

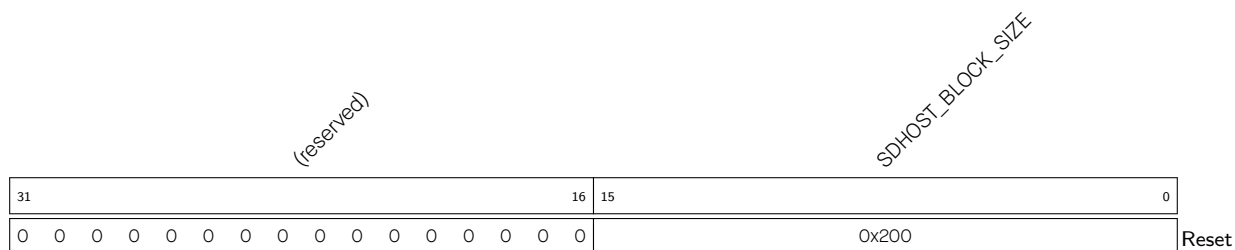
NOTE: The software timer should be used if the timeout value is in the order of 100 ms. In this case, read data timeout interrupt (DRT0) needs to be disabled.
(R/W)

Register 54.6. SDHOST_CTYPE_REG (0x0018)

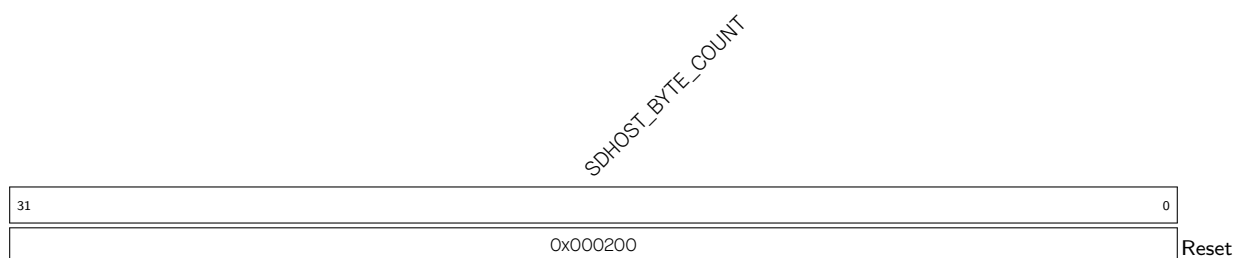


SDHOST_CARD_WIDTH4 Configures whether the card is 1-bit or 4-bit mode. One bit per card. Bit[0] corresponds to card[0]. For every card's bit:
0: 1-bit mode
1: 4-bit mode
(R/W)

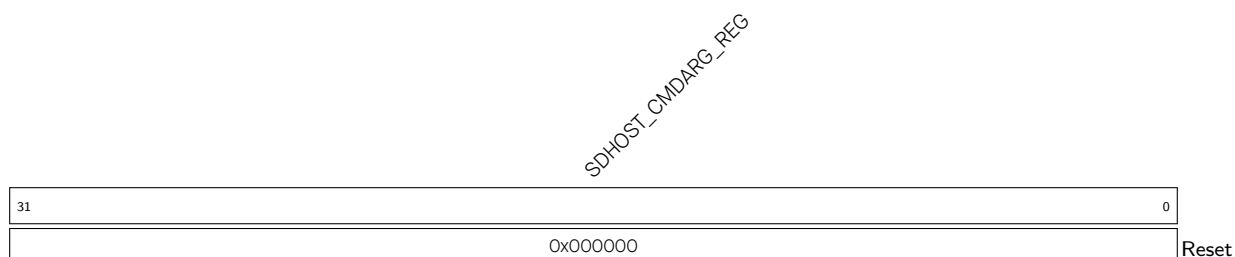
SDHOST_CARD_WIDTH8 Configures whether the card is 8-bit mode. One bit per card. Bit[0] corresponds to card[0]. For every card's bit:
0: Non 8-bit mode
1: 8-bit mode
(R/W)

Register 54.7. SDHOST_BLKSIZE_REG (0x001C)

SDHOST_BLOCK_SIZE Configures the Block Size. (R/W)

Register 54.8. SDHOST_BYTCNT_REG (0x0020)

SDHOST_BYTE_COUNT Configures the number of bytes to be transferred, should be an integral multiple of Block Size for block transfers. For data transfers of undefined byte lengths, the byte count should be set to 0. When the byte count is set to 0, it is the responsibility of the host to explicitly send a stop/abort command to terminate data transfer. (R/W)

Register 54.9. SDHOST_CMDARG_REG (0x0028)

SDHOST_CMDARG_REG Configures the value indicates the command argument to be passed to the card. (R/W)

Register 54.10. SDHOST_CMD_REG (0x002C)

SDHOST_START_CMD (reserved)				SDHOST_USE_HOLE_REG SDHOST_VOLT_SWITCH (reserved)				SDHOST_CCS_EXPECTED SDHOST_READ_CEATA_DEVICE SDHOST_UPDATE_CLOCK_REGISTERS_ONLY				SDHOST_CARD_NUMBER				SDHOST_SEND_INITIALIZATION SDHOST_STOP_ABORT_CMD SDHOST_WAIT_PRIVDATA_COMPLETE SDHOST_SEND_AUTO_STOP SDHOST_TRANSFER_MODE SDHOST_READ_WRITE SDHOST_DATA_EXPECTED SDHOST_CHECK_RESPONSE CRC SDHOST_RESPONSE_LENGTH SDHOST_RESPONSE_EXPECT				SDHOST_CMD_INDEX					
31	30	29	28	27	24		23	22	21	20	16		15	14	13	12	11	10	9	8	7	6	5	0	
0	0	1	0	0	0	0	0	0	0	0	0x0		0	0	0	0	0	0	0	0	0	0	0x0		Reset

SDHOST_CMD_INDEX Configures the Command index send to card. (R/W)

SDHOST_RESPONSE_EXPECT Configures whether to expect response from card.

0: No response expected from card

1: Response expected from card

(R/W)

SDHOST_RESPONSE_LENGTH Configures response length from card.

0: Short response expected from card

1: Long response expected from card

(R/W)

SDHOST_CHECK_RESPONSE_CRC Configures whether to check response CRC.

0: Do not check

1: Check response CRC

(R/W)

SDHOST_DATA_EXPECTED Configures whether to expect data transfer.

0: No data transfer expected

1: Data transfer expected

(R/W)

SDHOST_READ_WRITE Configures data transfer direction.

0: Read from card

1: Write to card

(R/W)

SDHOST_TRANSFER_MODE Configures data transfer mode.

0: Block data transfer command

1: Stream data transfer command

(R/W)

SDHOST_SEND_AUTO_STOP Configures whether to automatically send stop command.

0: No stop command is sent at the end of data transfer

1: Send stop command at the end of data transfer

(R/W)

Continued on the next page...

Register 54.10. SDHOST_CMD_REG (0x002C)

Continued from the previous page...

SDHOST_WAIT_PRVDATA_COMPLETE Configures whether to send command at once.

0: Send command at once, even if previous data transfer has not completed

1: Wait for previous data transfer to complete before sending Command

(R/W)

SDHOST_STOP_ABORT_CMD Configures whether the stop/abort command stops data transfer.

0: Neither stop nor abort command can stop the current data transfer. If abort is sent to the card device currently selected, or the card device is not in data-transfer mode, then bit should be set to 0

1: Stop or abort command intended to stop current data transfer in progress

(R/W)

SDHOST_SEND_INITIALIZATION Configures whether to send initialization sequence (80 clocks of 1) before sending this command. After powered on, 80 clocks must be sent to card for initialization before sending any commands to card. Bit should be set while sending the first command to card so that the controller will initialize clocks before sending command to card.

0: Do not send initialization sequence before sending this command

1: Send initialization sequence before sending this command

(R/W)

SDHOST_CARD_NUMBER Configures card number in use, the physical slot number of the card being accessed. In SD-only mode, up to two cards are supported. (R/W)

SDHOST_UPDATE_CLOCK_REGISTERS_ONLY Configures whether to only update clock register value into card clock domain.

Following register values are transferred into card clock domain: SDHOST_CLKDIV_REG, SDHOST_CLRSRC_REG, and SDHOST_CLKENA_REG.

When this bit is set, there will be no Command Done interrupts because no command is sent to SD/MMC/CEATA cards.

0: Normal command sequence

1: Do not send commands, just update clock register value into card clock domain

(R/W)

SDHOST_READ_CEATA_DEVICE Configures whether to read access CE-ATA device.

This bit is used to disable read data timeout indication while performing CE-ATA read transfers.

0: Host is not performing read access (RW_REG or RW_BLK) towards CE-ATA device

1: Host is performing read access (RW_REG or RW_BLK) towards CE-ATA device

(R/W)

Continued on the next page...

Register 54.10. SDHOST_CMD_REG (0x002C)

Continued from the previous page...

SDHOST_CCS_EXPECTED Configures whether to expect Command Completion Signal (CCS).

0: Interrupts are not enabled in CE-ATA device (nIEN = 1 in ATA control register), or command does not expect CCS from CE-ATA device

1: Interrupts are enabled in CE-ATA device (nIEN = 0), and RW_BLK command expects CCS from CE-ATA device

(R/W)

SDHOST_VOLT_SWITCH Configures whether to enable voltage switching.

0: No voltage switching

1: Voltage switching enabled, must be set for CMD11 only

(R/W)

SDHOST_USE_HOLE_REG Configures whether to use Hold Register.

0: CMD and DATA sent to card bypassing HOLD Register

1: CMD and DATA sent to card through the HOLD Register

(R/W)

SDHOST_START_CMD Configures whether to start the command.

0: Not start

1: Start

Once command is served by the CIU, this bit is automatically cleared.

When this bit is set, host should not attempt to write to any command registers. If a write is attempted, hardware lock error is set in raw interrupt register. (R/W/SC)

Register 54.11. SDHOST_FIFOTH_REG (0x004C)

(reserved)				SDHOST_DMA_MULTIPLE_TRANSACTION_SIZE				(reserved)				SDHOST_RX_WMARK				(reserved)				SDHOST_TX_WMARK			
31	30	28	27	26			16	15		12	11												0
0		0x0	0			0x0		0	0	0	0					0x0							Reset

SDHOST_TX_WMARK Configures FIFO threshold watermark level when transmitting data to card.

When FIFO data count is less than or equal to this number, DMA/FIFO request is raised. During end of packet, request or interrupt is generated, regardless of threshold programming.

In non-DMA mode, when transmitting FIFO threshold (TXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, on last interrupt, host is responsible for filling FIFO with only the required remaining bytes (not before FIFO is full or after CIU completes data transfers, because FIFO may not be empty).

In DMA mode, at end of packet, if last transfer is less than burst size, DMA controller does single cycles until required bytes are transferred. (R/W)

SDHOST_RX_WMARK Configures FIFO threshold watermark level when receiving data from card.

When FIFO data count reaches greater than this number, DMA/FIFO request is raised. During end of packet, request is generated regardless of threshold programming in order to complete any remaining data.

In non-DMA mode, when receiver FIFO threshold (RXDR) interrupt is enabled, then interrupt is generated instead of DMA request. During end of packet, interrupt is not generated if threshold programming is larger than any remaining data. It is responsibility of host to read remaining bytes on seeing Data Transfer Done interrupt.

In DMA mode, at end of packet, even if remaining bytes are less than threshold, DMA request does single transfers to flush out any remaining bytes before Data Transfer Done interrupt is set. (R/W)

SDHOST_DMA_MULTIPLE_TRANSACTION_SIZE Configures burst size of multiple transaction, should be programmed same as DMA controller multiple-transaction-size SDHOST_SRC/DEST_MSIZ.

0x0: 1-byte transfer

0x1: 4-byte transfer

0x2: 8-byte transfer

0x3: 16-byte transfer

0x4: 32-byte transfer

0x5: 64-byte transfer

0x6: 128-byte transfer

0x7: 256-byte transfer

(R/W)

Register 54.15. SDHOST_HCON_REG (0x0070)

(reserved)		(reserved)		SDHOST_NUM_CLK_DIV_REG		(reserved)		SDHOST_HOLD_REG		SDHOST_RAM_INDISE_REG		SDHOST_DMA_WIDTH_REG		(reserved)		SDHOST_ADDR_WIDTH_REG		SDHOST_DATA_WIDTH_REG		SDHOST_BUS_TYPE_REG		SDHOST_CARD_NUM_REG		SDHOST_CARD_TYPE_REG	
31	27	26	25	24	23	22	21	20	18	17	16	15	10	9	7	6	5	1	0						
0x0		0x0	0x3	0x1	0x1	0x0	0x1		0x0	0x13			0x1		0x1	0x1		0x1	Reset						

SDHOST_NUM_CLK_DIV_REG Represents 4 clk divider in design. (RO)

SDHOST_HOLD_REG Represents a hold register in data path. (RO)

SDHOST_RAM_INDISE_REG Represents that the SDMMC module contains RAM. (RO)

SDHOST DMA WIDTH REG Represents that the DMA data width is 32. (RO)

SDHOST_ADDR_WIDTH_REG Represents that the register address width is 32. (RO)

SDHOST_DATA_WIDTH_REG Represents that the register data width is 32. (RO)

SDHOST_BUS_TYPE_REG Represents that the register is configured for the APB bus. (RO)

SDHOST_CARD_NUM_REG Represents that the number of supported cards is 2. (RO)

SDHOST_CARD_TYPE_REG Represents that the hardware supports SDIO and MMC. (RO)

Register 54.16. SDHOST_UHS_REG (0x0074)

[illegible]

SDHOST_VOLT_REG Configures High Voltage mode per card. Bit[0] corresponds to card[0]. For every card bit:

- 0: 3.3 V VDD
- 1: 1.8 V VDD

(R/W)

SDHOST_DDR_REG Configures DDR (double data rate) mode selection per card. Bit[0] corresponds to card[0]. For every card bit:

- 0: Non-DDR mode
- 1: DDR mode

(R/W)

Register 54.17. SDHOST_RST_N_REG (0x0078)

(reserved)																															SDHOST_CARD_RESET			
31																															2	1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0x1	Reset	

SDHOST_CARD_RESET Configures hardware reset per card. Bit[0] corresponds to card[0]. These bits cause the cards to enter the pre-idle state, which requires them to be re-initialized. For every card bit:

- 0: Reset
- 1: Active mode

(R/W)

Register 54.18. SDHOST_BMOD_REG (0x0080)

(reserved)											SDHOST_BMOD_PBL		SDHOST_BMOD_DE		(reserved)		SDHOST_BMOD_FB		SDHOST_BMOD_SWR	
31											11	10	8	7	6			2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset

SDHOST_BMOD_SWR Configures whether to reset all DMA internal registers. It is automatically cleared after one clock cycle.

0: No effect

1: Reset

(R/W)

SDHOST_BMOD_FB Configures whether the AHB Master interface performs fixed burst transfers.

0: Use SINGLE and INCR burst transfer operations

1: Only use SINGLE, INCR4, INCR8 or INCR16 burst transfers

(R/W)

SDHOST_BMOD_DE Configures whether to enable the DMA.

0: Not enable

1: Enable

(R/W)

SDHOST_BMOD_PBL Configures the maximum number of beats to be performed in one DMA transaction.

0x0: 1-beat transfer

0x1: 4-beat transfer

0x2: 8-beat transfer

0x3: 16-beat transfer

0x4: 32-beat transfer

0x5: 64-beat transfer

0x6: 128-beat transfer

0x7: 256-beat transfer

(R/W)

Register 54.19. SDHOST_PLDMND_REG (0x0084)

SDHOST_PLDMND_PD	
31	0
0x000000	
Reset	

SDHOST_PLDMND_PD Configures the DMA FSM to resume normal descriptor fetch operation. If the OWNER bit of a descriptor is not set, the FSM goes to the Suspend state, software needs to write any value into this field for the FSM to resume normal descriptor fetch operation. (WO)

Register 54.20. SDHOST_DBADDR_REG (0x0088)

SDHOST_DBADDR_REG	
31	0
0x000000	
Reset	

SDHOST_DBADDR_REG Configures the base address of the linked list (First Descriptor). Bit[1:0] are ignored and taken as all-zero by the DMA internally. (R/W)

Register 54.21. SDHOST_CARDTHRCTL_REG (0x0100)

SDHOST_CARDTHRESHOLD_REG																(reserved)																SDHOST_CARDOLDF		SDHOST_CARDNEW	
31																15																2		1	0
0x00																0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0																0		0	Reset

SDHOST_CARDRDTHREN_REG Configures whether to enable card read threshold.

0: Not enable

1: Enable

(R/W)

SDHOST_CARDCLRINTEN_REG Configures whether to enable busy clear interrupt generation.

0: Not enable

1: Enable

(R/W)

SDHOST_CARDTHRESHOLD_REG Configures the card read threshold. Measurement unit: byte.

This field is valid only when SDHOST_CARDRDTHREN_REG is set to 1.

The value should be less than the FIFO size 512.(R/W)

Register 54.22. SDHOST_EMMCDDR_REG (0x010C)

Diagram illustrating the structure of the `SDHOST_HALFS` register. The register is 32 bits wide, with bits 31 down to 0. Bits 31-2 are reserved. Bit 1 is the `SDHOST_HALFS` bit. Bit 0 is the `Reset` bit.

SDHOST_HALFSTARTBIT_REG Configures the start bit detection mechanism duration of start bit.

Each bit refers to one card. Set this bit to 1 for eMMC4.5 and above, set to 0 for SD applications.

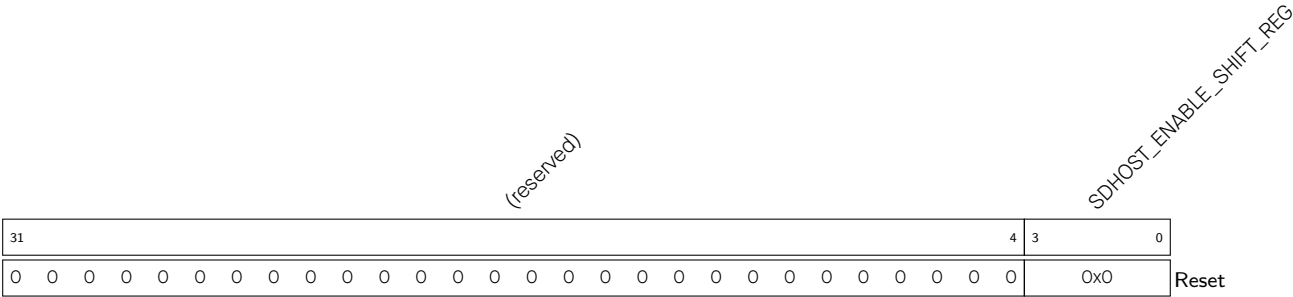
For eMMC4.5, start bit can be:

0: Full cycle

1: Less than one full cycle

(R/W)

Register 54.23. SDHOST_ENSHIFT_REG (0x0110)

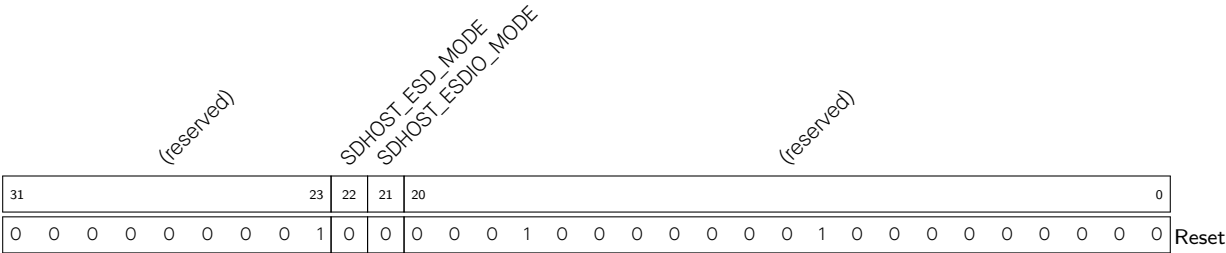


SDHOST_ENABLE_SHIFT_REG Configures the amount of phase shift provided on the default enables in the design. Two bits are assigned for each card, e.g., bit[1:0] are assigned to card 0. For every card:

- 0x0: Default phase shift
- 0x1: Enables shifted to next immediate positive edge
- 0x2: Enables shifted to next immediate negative edge
- 0x3: Reserved

(R/W)

Register 54.24. SDHOST_CLK_EDGE_SEL_REG (0x0800)



SDHOST_ESDIO_MODE Configures whether to enable eSDIO mode.

- 0: Not enable
- 1: Enable

(R/W)

SDHOST_ESD_MODE Configures whether to enable eSD mode.

- 0: Not enable
- 1: Enable

(R/W)

(reserved)

SDHOST_DLL_CCLK_IN_SAM_PHASE

SDHOST_DLL_CCLK_IN_DRV_PHASE

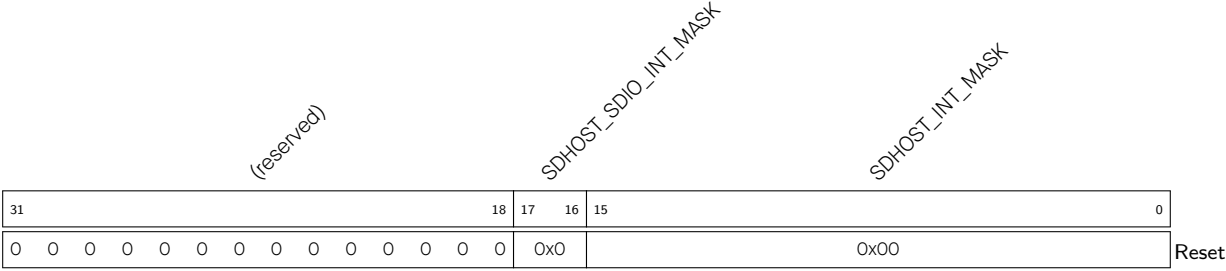
SDHOST_DLL_CCLK_IN_SLF_PHASE (reserved)

(reserved)

[illegible]

SDHOST_DLL_CCLK_IN_SAM_PHASE Configures the clock phase of the input signal when selecting high frequency clock. Unit is 1/64 of clock cycle. (R/W)

Register 54.26. SDHOST_INTMASK_REG (0x0024)

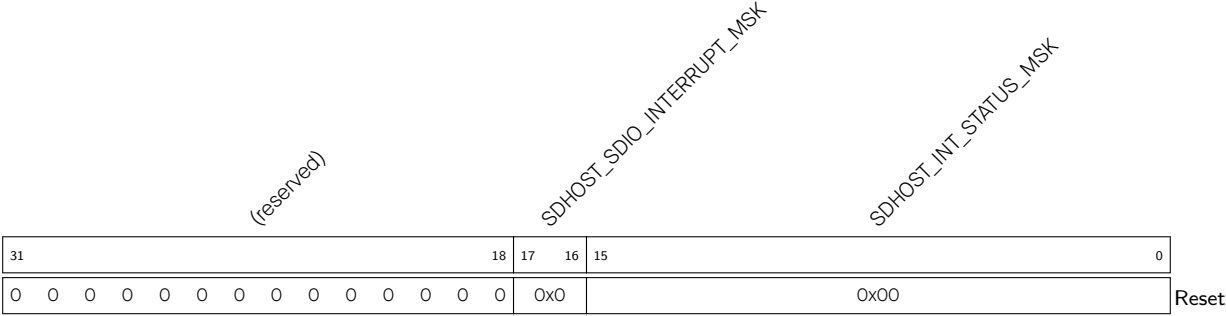


SDHOST_INT_MASK Write 1 to correspond bit to enable correspond interrupt. For every bit:

- Bit 15 (EBE): End-bit error/no CRC error
 - Bit 14 (ACD): Auto command done
 - Bit 13 (SBE/BCI): Rx Start Bit Error
 - Bit 12 (HLE): Hardware locked write error
 - Bit 11 (FRUN): FIFO underrun/overflow error
 - Bit 10 (HTO): Data starvation-by-host timeout
 - Bit 9 (DRT0): Data read timeout
 - Bit 8 (RTO): Response timeout
 - Bit 7 (DCRC): Data CRC error
 - Bit 6 (RCRC): Response CRC error
 - Bit 5 (RXDR): Receive FIFO data request
 - Bit 4 (TXDR): Transmit FIFO data request
 - Bit 3 (DTO): Data transfer over
 - Bit 2 (CMDD): Command done
 - Bit 1 (RE): Response error
 - Bit 0 (CD): Card detect
- (R/W)

SDHOST_SDIO_INT_MASK Write 1 to correspond bit to enable interrupt from correspond card. One bit per card. Bit[0] corresponds to card[0]. (R/W)

Register 54.27. SDHOST_MINTSTS_REG (0x0040)

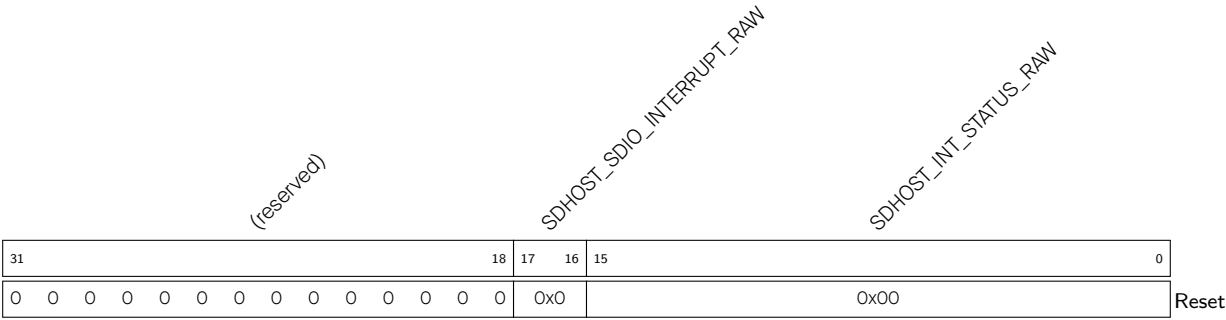


SDHOST_INT_STATUS_MSK The masked interrupt status. For every bit:

- Bit 15 (EBE): End-bit error/no CRC error
- Bit 14 (ACD): Auto command done
- Bit 13 (SBE/BCI): RX Start Bit Error
- Bit 12 (HLE): Hardware locked write error
- Bit 11 (FRUN): FIFO underrun/overflow error
- Bit 10 (HTO): Data starvation by host timeout
- Bit 9 (DTRO): Data read timeout
- Bit 8 (RTO): Response timeout
- Bit 7 (DCRC): Data CRC error
- Bit 6 (RCRC): Response CRC error
- Bit 5 (RXDR): Receive FIFO data request
- Bit 4 (TXDR): Transmit FIFO data request
- Bit 3 (DTO): Data transfer over
- Bit 2 (CMDD): Command done
- Bit 1 (RE): Response error
- Bit 0 (CD): Card detect (RO)

SDHOST_SDIO_INTERRUPT_MSK The masked interrupt status from SDIO card. One bit for each card. Bit[0] corresponds to card[0]. (RO)

Register 54.28. SDHOST_RINTSTS_REG (0x0044)



SDHOST_INT_STATUS_RAW The raw interrupt status. For every bit:

- Bit 15 (EBE): End-bit error/no CRC error
- Bit 14 (ACD): Auto command done
- Bit 13 (SBE/BCI): RX Start Bit Error
- Bit 12 (HLE): Hardware locked write error
- Bit 11 (FRUN): FIFO underrun/overflow error
- Bit 10 (HTO): Data starvation by host timeout
- Bit 9 (DTRO): Data read timeout
- Bit 8 (RTO): Response timeout
- Bit 7 (DCRC): Data CRC error
- Bit 6 (RCRC): Response CRC error
- Bit 5 (RXDR): Receive FIFO data request
- Bit 4 (TXDR): Transmit FIFO data request
- Bit 3 (DTO): Data transfer over
- Bit 2 (CMDD): Command done
- Bit 1 (RE): Response error
- Bit 0 (CD): Card detect (R/W1C)

SDHOST_SDIO_INTERRUPT_RAW The raw interrupt status from SDIO card. One bit per card. Bit[0] corresponds to card[0]. (R/W1C)

Register 54.29. SDHOST_IDSTS_REG (0x008C)

[illegible]

SDHOST_IDSTS_TI The raw interrupt status of Transmit Interrupt. Indicates that data transmission is finished for a descriptor. (R/W1C)

SDHOST_IDSTS_RI The raw interrupt status of Receive Interrupt. Indicates the completion of data reception for a descriptor. (R/W1C)

SDHOST_IDSTS_FBE The raw interrupt status of Fatal Bus Error Interrupt. Indicates that a Bus Error occurred (SDHOST_IDSTS_REG[12:10]). (R/W1C)

SDHOST_IDSTS_DU The raw interrupt status of Descriptor Unavailable Interrupt. This bit is set when the descriptor is unavailable due to OWNER bit = 0 (DESO[31] = 0). (R/W1C)

SDHOST_IDSTS_CES The raw interrupt status of Card Error Summary. Indicates the status of the transaction to/from the card, also present in SDHOST_RINTSTS_REG. Indicates the logical OR of the following bits:

EBE: End-bit error/no CRC error

SBE/BCI: RX Start Bit Error

DRT0: Data read timeout

RTO: Response timeout

DCRC: Data CRC error

RCRC: Response CRC error

RE: Response error

The abort condition of the DMA depends on the setting of this field. If this field is enabled, then the DMA aborts on a response error. (R/W1C)

SDHOST_IDSTS_NIS The raw interrupt status of Normal Interrupt Summary. Logical OR of SDHOST_IDSTS_REG[1:0]. Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes this field to be set is cleared by software. (R/WIC)

SDHOST_IDSTS_AIS The raw interrupt status of Abnormal Interrupt Summary. Logical OR of SDHOST_IDSTS_REG[2], SDHOST_IDSTS_REG[4]. Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes this field to be set is cleared by software. (R/W1C)

Continued on the next page...

Register 54.29. SDHOST_IDSTS_REG (0x008C)

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SDHOST_IDSTS_FBE_CODE Represents the type of error that caused a Bus Error, valid only when SDHOST_IDSTS_REG[2] is set.

0x1: Host Abort received during transmission

0x2: Host Abort received during reception

Others: Reserved

(RO)

SDHOST_IDSTS_FSM Represents DMA FSM present state.

0x0: DMA_IDLE (idle state)

0x1: DMA_SUSPEND (suspend state)

0x2: DESC_RD (descriptor reading state)

0x3: DESC_CHK (descriptor checking state)

0x4: DMA_RD_REQ_WAIT (read-data request waiting state)

0x5: DMA_WR_REQ_WAIT (write-data request waiting state)

0x6: DMA_RD (data-read state)

0x7: DMA_WR (data-write state)

0x8: DESC_CLOSE (descriptor close state)

(RO)

Register 54.30. SDHOST_IDINTEN_REG (0x0090)

(reserved)																						SDHOST_IDINTEN_AI		SDHOST_IDINTEN_NI		(reserved)		SDHOST_IDINTEN_CES		SDHOST_IDINTEN_DU		SDHOST_IDINTEN_FBE		SDHOST_IDINTEN_RI		SDHOST_IDINTEN_TI	
31																						10		9	8	7	6	5	4	3	2	1	0				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset						

SDHOST_IDINTEN_TI Write 1 to enable interrupt of Transmit. (R/W)

SDHOST_IDINTEN_RI Write 1 to enable interrupt of Receive. (R/W)

SDHOST_IDINTEN_FBE Write 1 to enable interrupt of Fatal Bus Error. (R/W)

SDHOST_IDINTEN_DU Write 1 to enable interrupt of Descriptor Unavailable. (R/W)

SDHOST_IDINTEN_CES Write 1 to enable interrupt of Card Error summary. (R/W)

SDHOST_IDINTEN_NI Write 1 to enable interrupt of Normal Interrupt Summary. (R/W)

SDHOST_IDINTEN_AI Write 1 to enable interrupt of Abnormal Interrupt Summary. (R/W)

Register 54.31. SDHOST_RESP0_REG (0x0030)

SDHOST_RESPONSE0_REG

31	0
0x000000	
Reset	

SDHOST_RESPONSE0_REG Represents bit[31:0] of response. (RO)

Register 54.32. SDHOST_RESP1_REG (0x0034)

SDHOST_RESPONSE1_REG

31	0
0x000000	
Reset	

SDHOST_RESPONSE1_REG Represents bit[63:32] of long response. (RO)

Register 54.33. SDHOST_RESP2_REG (0x0038)

SDHOST_RESPONSE2_REG

31	0
0x000000	
Reset	

SDHOST_RESPONSE2_REG Represents bit[95:64] of long response. (RO)

Register 54.34. SDHOST_RESP3_REG (0x003C)

SDHOST_RESPONSE3_REG																															
31																															0
0x000000																															
Reset																															

SDHOST_RESPONSE3_REG Represents bit[127:96] of long response. (RO)

Register 54.35. SDHOST_STATUS_REG (0x0048)

(reserved)																															SDHOST_FIF0_COUNT																SDHOST_RESPONSE_INDEX										SDHOST_DATA_STATE_MC_BUSY										SDHOST_DATA_BUSY										SDHOST_DATA_3_STATUS										SDHOST_COMMAND_FSM_STATES										SDHOST_FIF0_FULL										SDHOST_FIF0_EMPTY										SDHOST_FIF0_TX_WATERMARK										SDHOST_FIF0_RX_WATERMARK									
31	30	29																	17	16											11	10	9	8	7											4	3	2	1	0																																																																																						
0	0	0x00																0x0										1	1	1	0x1										0	1	1	0	Reset																																																																																											

SDHOST_FIFO_RX_WATERMARK Represents whether the FIFO reached Receive watermark level, not qualified with data transfer.
 0: Not reach
 1: Reach
 (RO)

SDHOST_FIFO_TX_WATERMARK Represents whether the FIFO reached Transmit watermark level, not qualified with data transfer.
 0: Not reach
 1: Reach
 (RO)

SDHOST_FIFO_EMPTY Represents whether the FIFO is empty.
 0: Not empty
 1: Empty
 (RO)

SDHOST_FIFO_FULL Represents whether the FIFO is full.
 0: Not full
 1: Full
 (RO)

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Register 54.35. SDHOST_STATUS_REG (0x0048)

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SDHOST_COMMAND_FSM_STATES Represents the command FSM states.

0x0: Idle
0x1: Send init sequence
0x2: Send cmd start bit
0x3: Send cmd tx bit
0x4: Send cmd index + arg
0x5: Send cmd crc7
0x6: Send cmd end bit
0x7: Receive resp start bit
0x8: Receive resp IRQ response
0x9: Receive resp tx bit
0xA: Receive resp cmd idx
0xB: Receive resp data
0xC: Receive resp crc7
0xD: Receive resp end bit
0xE: Cmd path wait NCC
0xF: Wait, cmd-to-response turnaround
(RO)

SDHOST_DATA_3_STATUS Represents the value of the sdhost_card_data[3] signal, which indicates whether card is present.

0: Card not present
1: Card present
(RO)

SDHOST_DATA_BUSY Represents inverted value of the sdhost_card_data[0] signal, which indicates card data is busy.

0: Card data not busy
1: Card data busy
(RO)

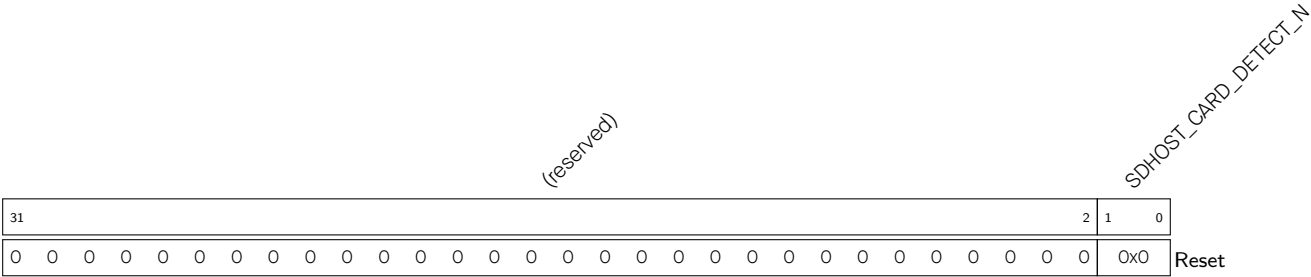
SDHOST_DATA_STATE_MC_BUSY Represents whether the data transmit or receive state-machine is busy.

0: Not busy
1: Busy
(RO)

SDHOST_RESPONSE_INDEX Represents index of previous response, including any auto-stop sent by core. (RO)

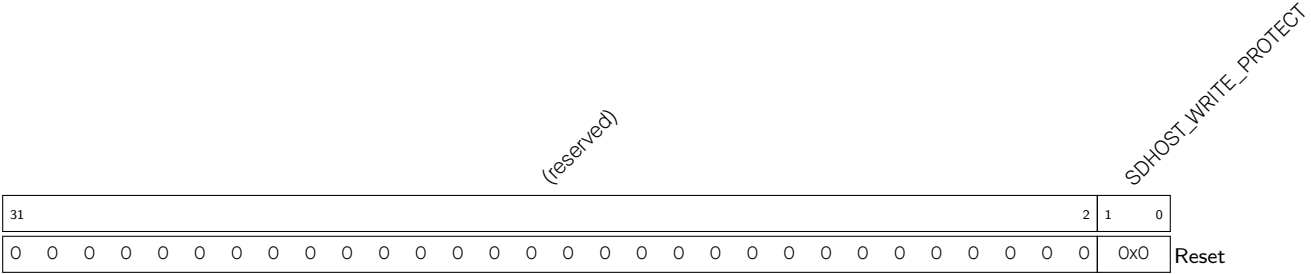
SDHOST_FIFO_COUNT Represents FIFO count, number of filled locations in FIFO. (RO)

Register 54.36. SDHOST_CDETECT_REG (0x0050)



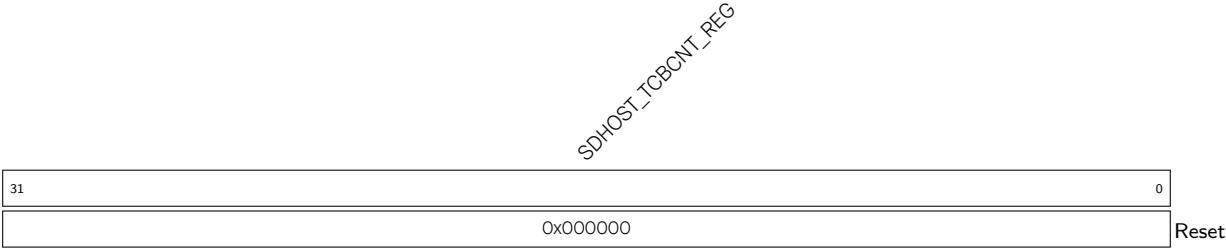
SDHOST_CARD_DETECT_N Represents value on sdhost_card_detect_n input ports. One bit per card. Bit[0] corresponds to card[0]. (RO)

Register 54.37. SDHOST_WRTprt_REG (0x0054)

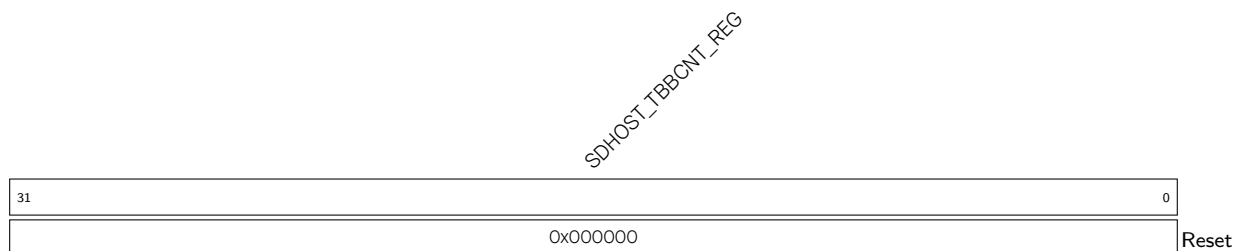


SDHOST_WRITE_PROTECT Represents value on sdhost_card_write_prt input ports, one bit per card. Bit[0] corresponds to card[0]. (RO)

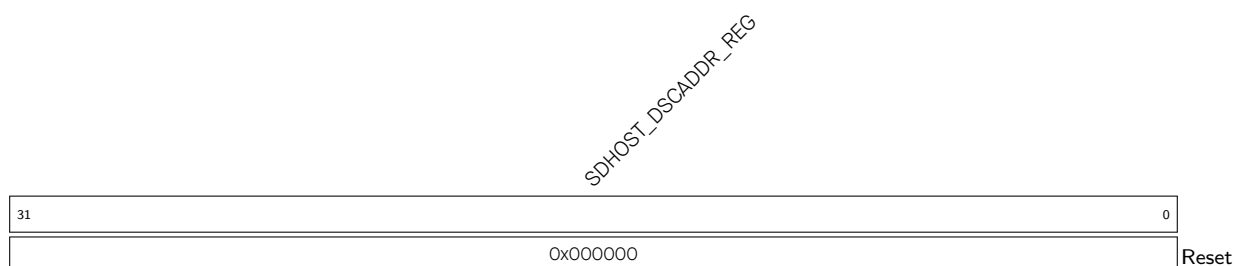
Register 54.38. SDHOST_TCBCNT_REG (0x005C)



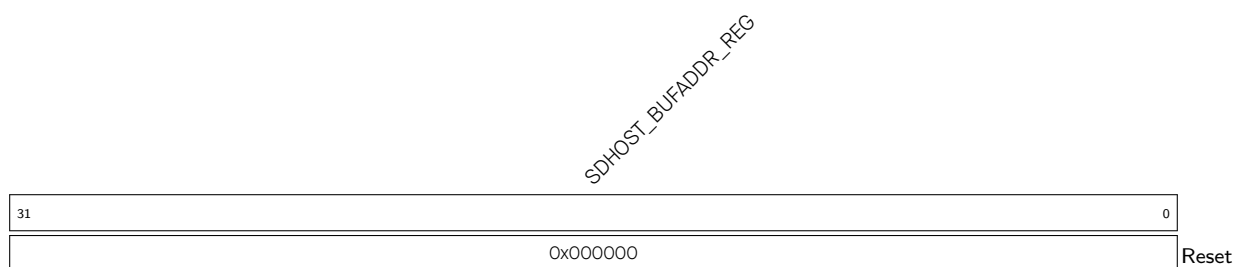
SDHOST_TCBCNT_REG Represents the number of bytes transferred by the CIU unit to the card. (RO)

Register 54.39. SDHOST_TBBCNT_REG (0x0060)

SDHOST_TBBCNT_REG Represents the number of bytes transferred between Host/DMA memory and BIU FIFO. (RO)

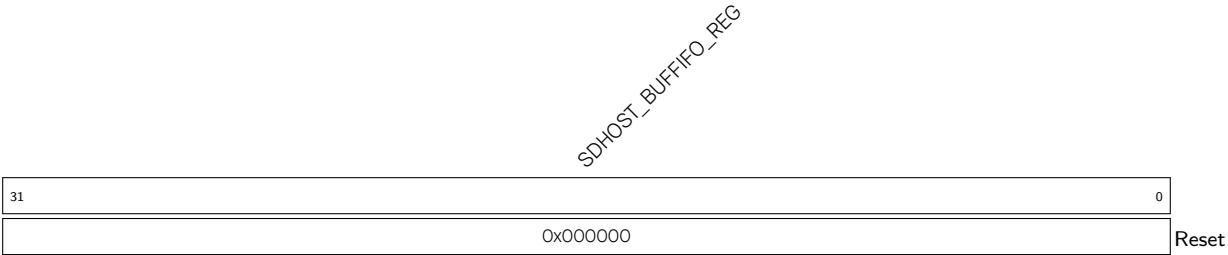
Register 54.40. SDHOST_DSCADDR_REG (0x0094)

SDHOST_DSCADDR_REG Represents the Descriptor Address Pointer, points to the start address of current descriptor read by DMA. (RO)

Register 54.41. SDHOST_BUFADDR_REG (0x0098)

SDHOST_BUFADDR_REG Represents the Buffer Address Pointer, points to the current Data Buffer Address being accessed by the DMA. (RO)

Register 54.42. SDHOST_BUFFIFO_REG (0x0200)



SDHOST_BUFFIFO_REG Date received or to be transmitted between FIFO and software directly. This register points to the current Data FIFO. (R/W)